Back Propagation

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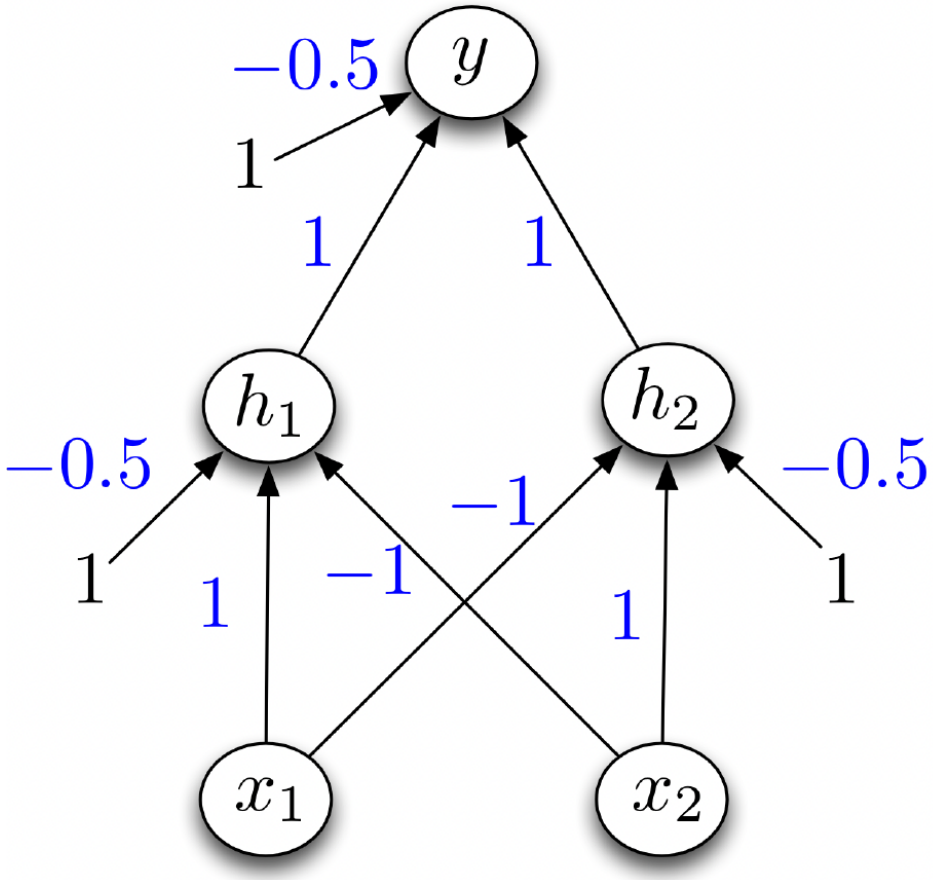
Main sources: slides of Stanley Chan

Non-linearity is Key in DL

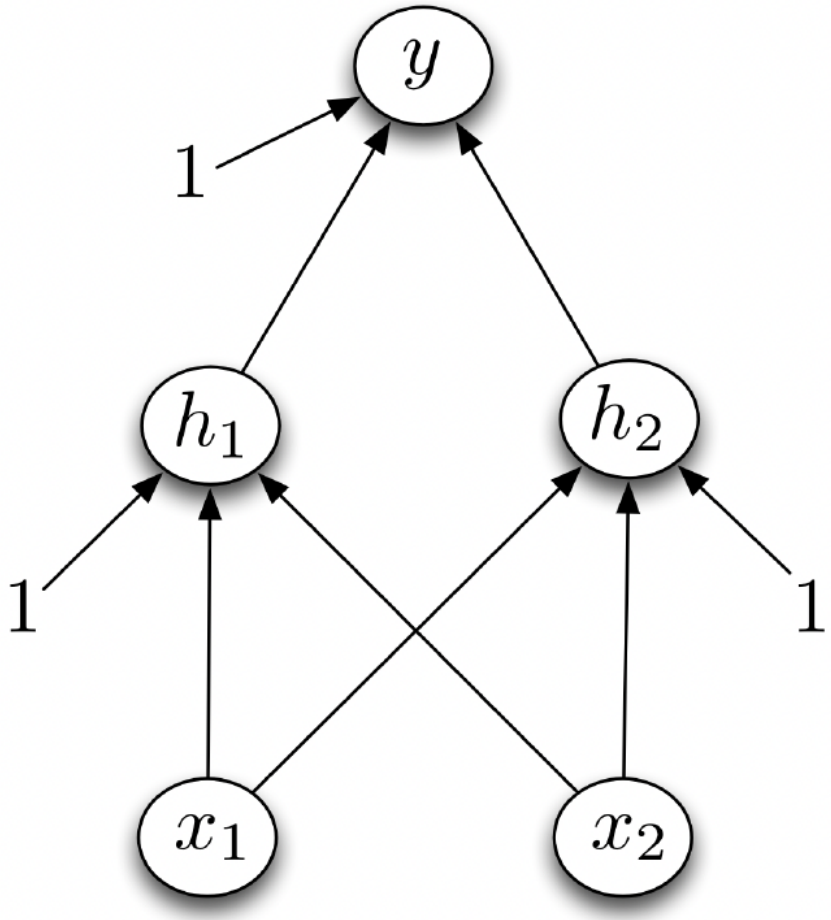
- Linear MLPs have limited potential
- Non-linearity through activation functions is key!

MLP for XOR

- Recall: linear MLP cannot model XOR perfectly
- Assume *hard threshold* activation function



How to learn the weights?



- Q: how to learn the weights?
  - What do we need?

What do we need?

- Data: $(\boldsymbol{x}_i, \boldsymbol{y}_i)_{i \in \mathcal{N}}$ pairs
- Loss function: a metric to measure how well the model performs

$$J(\boldsymbol{W}_1, \cdots, \boldsymbol{W}_L) = \sum_{i=1}^{N} \left\| \sigma\left(\mathbf{W}_L^T \ldots \sigma\left(\mathbf{W}_2^T \sigma\left(\mathbf{W}_1^T \mathbf{x}_i\right)\right)\right) - \mathbf{y}_i \right\|^2$$

  - $\sigma(\cdot)$: activation function, e.g., ReLU, Sigmoid
- Learning (optimization): gradient descent. E.g., for $\boldsymbol{W}_1$

$$\mathbf{W}_1^{t+1} = \mathbf{W}_1^t - \alpha \nabla J(\mathbf{W}_1^t)$$

  - A brief intro to gradient descent
  - α: learning rate
  - We need to do all for $\boldsymbol{W}_1, \cdots, \boldsymbol{W}_L$, brute force!

Example

- Consider two layers

$$J(\mathbf{W}_1, \mathbf{W}_2) = \left\| \underbrace{\sigma\left(\mathbf{W}_2^T \sigma\left(\mathbf{W}_1^T \mathbf{x}\right)\right)}_{\boldsymbol{a}_2} - \mathbf{y} \right\|^2$$

- We want to update both $\boldsymbol{W}_2$ and $\boldsymbol{W}_1$
- Let's start with $\boldsymbol{W}_2$ (backward)

$$\frac{\partial J}{\partial \mathbf{W}_2} = \frac{\partial J}{\partial a_2} \cdot \frac{\partial a_2}{\partial \mathbf{W}_2} \quad \text{(Chain Rule)}$$

$$a_2 = \sigma\left(\mathbf{W}_2^T \sigma\left(\mathbf{W}_1^T \mathbf{x}\right)\right)$$

$$\frac{\partial a_2}{\partial \mathbf{W}_2} = \frac{\partial a_2}{\partial z_2} \cdot \frac{\partial z_2}{\partial \mathbf{W}_2}$$

Example

$$J(\mathbf{W}_1, \mathbf{W}_2) = \left\| \sigma \left(\mathbf{W}_2^T \underbrace{\sigma \left(\mathbf{W}_1^T \mathbf{x} \right)}_{\boldsymbol{a}_1} \right) - \mathbf{y} \right\|^2$$

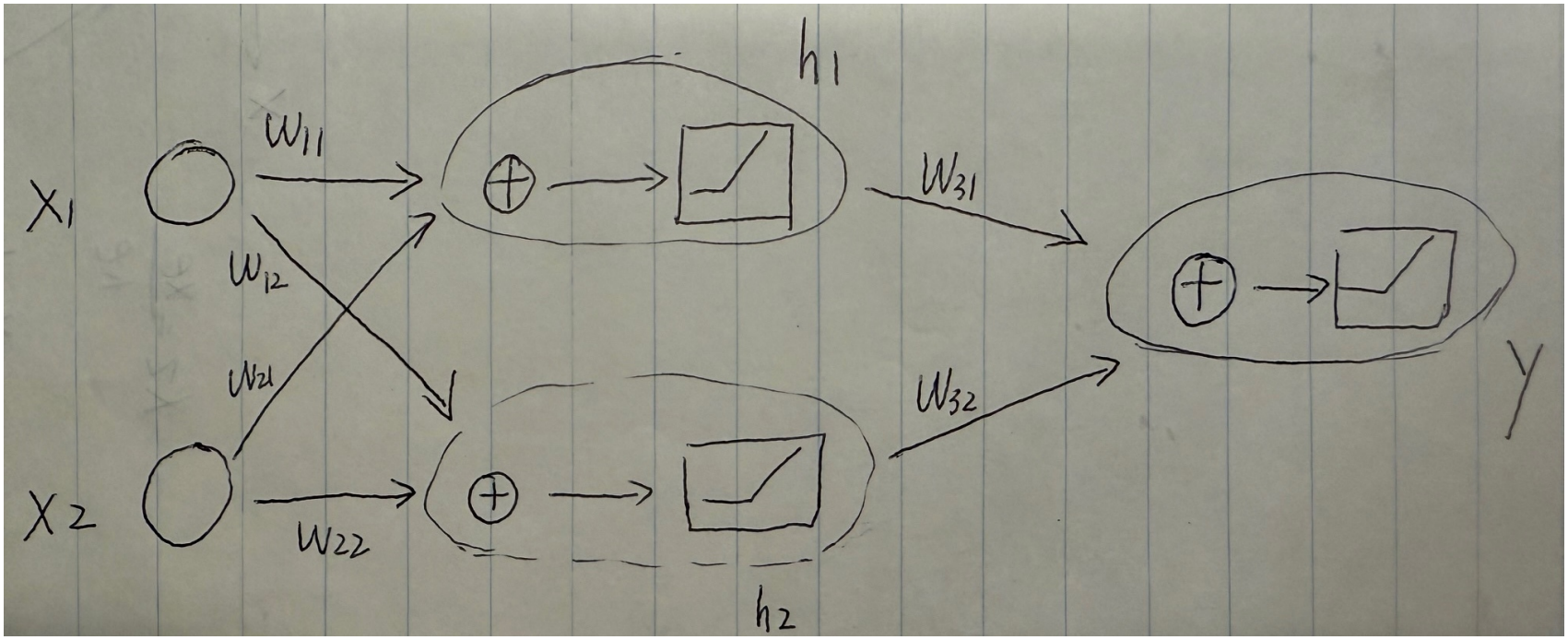
- What about $\boldsymbol{W}_1$?

$$\frac{\partial J}{\partial \mathbf{W}_1} = \frac{\partial J}{\partial a_2} \cdot \frac{\partial a_2}{\partial \mathbf{W}_1}$$

- Now we can repeat the calculation (Let $\boldsymbol{z}_1 = \boldsymbol{W}_1^T \boldsymbol{x}$)

$$\frac{\partial a_2}{\partial \mathbf{W}_1} = \frac{\partial a_2}{\partial a_1} \cdot \frac{\partial a_1}{\partial \mathbf{W}_1} = \frac{\partial a_2}{\partial a_1} \cdot \frac{\partial a_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial \mathbf{W}_1}$$

Numerical Example



- All weights initialized to 1 (random in practice)
- Learning rate: 0.01 (a key hyper-parameter in DL)
- Activation function: $\text{ReLu}(z) = \max\{0, z\}$; MSE as loss function
- Training data: $\boldsymbol{x} = (1, 1), y = 1$
- Task: use back propagation to update the weights

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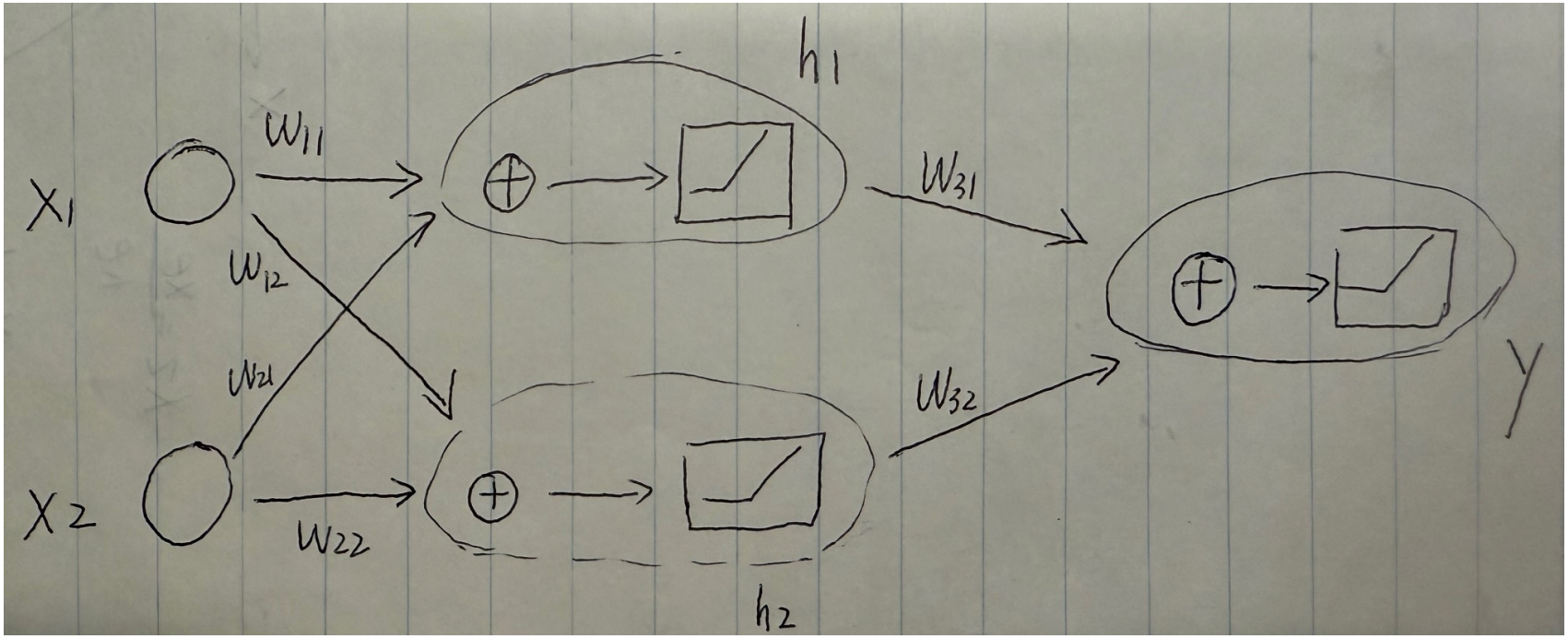

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Step 1: Forward Pass

Input to Hidden Layer

- Calculate z_1 and z_2:

$$z_1 = w_{11} \cdot x_1 + w_{21} \cdot x_2 = 1 \cdot 1 + 1 \cdot 1 = 2$$

$$z_2 = w_{12} \cdot x_1 + w_{22} \cdot x_2 = 1 \cdot 1 + 1 \cdot 1 = 2$$

- Apply ReLU Activation to get h_1 and h_2:

$$h_1 = \text{ReLU}(z_1) = \max(0, 2) = 2$$

$$h_2 = \text{ReLU}(z_2) = \max(0, 2) = 2$$

Step 1: Forward Pass

Hidden to Output Layer

- Calculate z_o:

$$z_o = w_{31} \cdot h_1 + w_{32} \cdot h_2 = 1 \cdot 2 + 1 \cdot 2 = 4$$

- Apply ReLU Activation to get o:

$$o = \text{ReLU}(z_o) = \max(0, 4) = 4$$

Initial Loss Calculation

$$J = \frac{1}{2}(o - y)^2 = \frac{1}{2}(4 - 1)^2 = \frac{1}{2} \cdot 9 = 4.5$$

Step 2: Backward Pass (Gradient Calculation)

- Goal: we need to update all weights, using gradient descent!
  - $\frac{\partial J}{\partial w_{31}}, \frac{\partial J}{\partial w_{32}}, \frac{\partial J}{\partial w_{11}}, \frac{\partial J}{\partial w_{12}}, \frac{\partial J}{\partial w_{21}}, \frac{\partial J}{\partial w_{22}}$
- **Gradient for Output Layer**, e.g., $\frac{\partial J}{\partial w_{31}}$

$$\frac{\partial J}{\partial w_{31}} = \frac{\partial J}{\partial z_o} \cdot \frac{\partial z_o}{\partial w_{31}}$$

  - $J = \frac{1}{2}(o - y)^2$, $o = \mathrm{ReLU}(z_0)$
  - $z_o = w_{31} h_1 + w_{32} h_2$

$$\frac{\partial J}{\partial z_o} = \frac{\partial J}{\partial o} \cdot \mathrm{ReLU}'(z_o) = 3 \cdot 1 = 3, \quad \frac{\partial z_o}{\partial w_{31}} = h_1 = 2$$

$$\frac{\partial J}{\partial w_{31}} = \frac{\partial J}{\partial z_o} \cdot h_1 = 3 \cdot 2 = 6$$

- Task: can you calculate $\frac{\partial J}{\partial w_{32}}$?

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Step 2: Backward Pass (Gradient Calculation)

- **Gradient for Hidden Layer**, e.g., $\frac{\partial J}{\partial w_{11}}$

$$\frac{\partial J}{\partial w_{11}} = \frac{\partial J}{\partial z_o} \cdot \frac{\partial z_o}{\partial h_1} \cdot \frac{\partial h_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_{11}}$$

  - This is a chain, isn't it?

$$\frac{\partial J}{\partial z_o} = 3, \quad \frac{\partial z_o}{\partial h_1} = w_{31} = 1; \quad \frac{\partial h_1}{\partial z_1} = 1; \quad \frac{\partial z_1}{\partial w_{11}} = x_1 = 1$$

$$\frac{\partial J}{\partial w_{11}} = 3$$

- Task: can you calculate $\frac{\partial J}{\partial w_{12}}, \frac{\partial J}{\partial w_{21}}, \frac{\partial J}{\partial w_{22}}$?

Step 2: Backward Pass (Gradient Calculation)

- **Gradient for Hidden Layer**, e.g., $\frac{\partial J}{\partial w_{11}}$

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$$\frac{\partial J}{\partial w_{11}} = 3$$

- Task: can you calculate $\frac{\partial J}{\partial w_{12}}$, $\frac{\partial J}{\partial w_{21}}$, $\frac{\partial J}{\partial w_{22}}$?

Step 3: Weight Update

- Update w_{31} and w_{32}:

$$w_{31} = w_{31} - \alpha \cdot \frac{\partial J}{\partial w_{31}} = 1 - 0.01 \cdot 6 = 1 - 0.06 = 0.94$$

$$w_{32} = w_{32} - \alpha \cdot \frac{\partial J}{\partial w_{32}} = 1 - 0.01 \cdot 6 = 1 - 0.06 = 0.94$$

- Update w_{11}, w_{12}, w_{21}, and w_{22}:

$$w_{11} = w_{11} - \alpha \cdot \frac{\partial J}{\partial w_{11}} = 1 - 0.01 \cdot 3 = 1 - 0.03 = 0.97$$

$$w_{12} = w_{12} - \alpha \cdot \frac{\partial J}{\partial w_{12}} = 1 - 0.01 \cdot 3 = 1 - 0.03 = 0.97$$

$$w_{21} = w_{21} - \alpha \cdot \frac{\partial J}{\partial w_{21}} = 1 - 0.01 \cdot 3 = 1 - 0.03 = 0.97$$

$$w_{22} = w_{22} - \alpha \cdot \frac{\partial J}{\partial w_{22}} = 1 - 0.01 \cdot 3 = 1 - 0.03 = 0.97$$

Quiz

- Given the updated weights, what would be the loss?
  - Use two decimals

Readings

- Ian Goodfellow and Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT press, 2016
  - e-Book available here: https://www.deeplearningbook.org/
  - Reading assignment: Chapter 6.5